**Course Code- 105406 Computer Network 3 0 0 3**

**Unit-1.0 4 hrs**

Overview of Data Communication and Networking: OSI Reference Model, TCP/IP Protocol Suite; Network Architecture and Physical Topology.

**Unit-2.0 10 hrs**

Physical Layer: Analog and Digital Signals, Transmission Impairment, Data Rate Limits, Performance Analysis of a Network; Representation and Synchronization of Bits, Analog and Digital Transmission; Multiplexing and Spreading Techniques; Guided Transmission Media; Circuit, Packet and Virtual Circuit Switching.

**Unit-3.0 10 hrs**

Data Link Layer: Framing, Flow and Error Control (Noiseless and Noisy Channels Protocols), PointToPoint Protocol; Random Access protocols (Pure/slotted ALOHA, CSMA/CD, CSMA/CA), Controlled Access Protocol (Bit-Map, Polling and Token Passing), Channelization (TDMA, FDMA, CDMA); Physical Addressing and Ethernet; Connecting LANs and Virtual LANs.

**Unit-4.0 7 hrs**

Network Layer: Internet Protocol version 4 and 6; Address Mapping (ARP, RARP, BOOTP and DHCP), ICMP and IGMP, Routing Algorithms.

**Unit-5.0 7 hrs**

Transport Layer: UDP, TCP; Congestion Control and QoS; Client-Server Model and Socket Interface.

**Unit-6.0 4 hrs**

Application Layer: DNS, Remote Logging, Electronic Mail (SMTP, POP), FTP, Introduction to WWW and HTTP.

**Text/ Reference:-**

1. B. Forouzan, “Data Communication and Network”, McGraw-Hill Publications. 4th ed.
2. A. S. Tanenbaum., “Computer Networks”, Pearson Education Asia. 5th ed.
3. W. Stalling, “Data and Computer Communication”, PHI (EEE). 8th ed.
4. A. L. Garcia and I. Widjaja, “Communication Networks: Fundamental Concepts and Key Architectures”, Tata McGraw-Hill. 2nd ed.
5. S. Sharma, “A course in Computer Networks”, Kataria. 3rd ed.